Alignment of Diffusion and Flow-Matching Models for Visual Generation: A Survey

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Abstract

Diffusion and flow-matching models have emerged as the dominant paradigms for scalable visual generation across image and video domains. As foundational models increase in generative capacity, research has shifted toward post-training alignment to better match human intent, improve controllability, and suppress undesirable behaviors. Aligning these models raises diffusion-native challenges, including reliance on imperfect feedback from learned reward models, and the difficulty of temporal credit assignment along long noise-dominated trajectories. This survey revisits the alignment literature from a diffusion-native perspective and utilizes a two-level taxonomy that distinguishes training-time and inference-time alignment, supported by a rigorous analysis of reward modeling paradigms. We further categorize training-time methods into five core paradigms: quality-aware fine-tuning, reward-weighted regression, reward feedback learning, policy-gradient optimization, and direct preference learning. By discussing key design choices and typical failure modes across these paradigms, we identify critical open challenges.

1 Introduction

Diffusion and flow-matching models have rapidly become the dominant paradigm for visual generation, enabling scalable synthesis across images [Esser *et al.*, 2024; Wu *et al.*, 2025a] and videos [Wan *et al.*, 2025]. As large-scale pretraining yields increasingly powerful foundational models, model quality and controllability are no longer limited solely by raw generative capacity, but also by which behaviors the model distribution prefers among many plausible generations. This shift has brought *post-training alignment* to the forefront: refining a pretrained generator to better match human intent and preferences, improving perceptual quality and controllability while suppressing undesirable outputs.

Alignment has been extensively studied in large language models (LLMs) and vision-language models (VLMs), and many conceptual tools naturally carry over to visual generators. However, diffusion and flow-matching generators also introduce **diffusion-native challenges** that make alignment substantially less straightforward than in the discrete-token setting. At first, visual alignment typically depends on *imperfect and non-verifiable feedback* like learned reward models [Xu *et al.*, 2023], whose reliability may degrade under distribution shift and may be exploited by over-optimization. Moreover, preferences are defined on the final decoded sample, whereas generation proceeds through long noisy trajectories [Black *et al.*, 2023], leaving *credit assignment across intermediate steps inherently difficult*.

Recent surveys [Wang *et al.*, 2024; Winata *et al.*, 2025] have reviewed alignment techniques broadly across model families, yet diffusion/flow post-training alignment has evolved rapidly and exhibits system-level differences. Motivated by this rapid growth, we revisit post-training alignment for diffusion and flow-matching models from a diffusion-native perspective and survey representative approaches across both training and inference. Figure 1 summarizes this trend by positioning representative reward models, training-time objectives, inference-time methods, and recent foundation models on a unified timeline. Throughout the survey, we adopt a unifying view that treats alignment as reshaping a generator's output distribution toward preferred samples while away from undesired ones. In diffusion and flow-matching models, such reshaping can be realized by modifying the denoising trajectory through parameter updates during training, or by steering sampling trajectories at inference time. The core technical question is how to translate imperfect preference feedback into effective and reliable control of diffusion and flow-matching generators, without sacrificing stability, diversity, or robustness under distribution shift.

This survey makes the following contributions: (i) We utilize a two-level taxonomy that distinguishes training-time alignment and inference-time alignment, and further organizes training-time methods into five representative paradigms, including QFT, RWR, ReFL, policy-gradient optimization, and DPL. (ii) We provide a detailed analysis of each paradigm, evaluating the feedback source, optimization objective, and trajectory intervention point, and we highlight their respective advantages in practice. (iii) We identify several open challenges, including reward reliability under distribution shift, reward hacking, and the opportunity of struc-

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tured environments with verifiable objectives.

2 Background

2.1 Diffusion and Flow-Matching Models

Generative models based on diffusion [Ho *et al.*, 2020] and flow-matching [Lipman *et al.*, 2022; Liu *et al.*, 2022] define a transformation from a noise distribution $x_T \sim \mathcal{N}(0, I)$ to a data distribution $x_0 \sim p_{\text{data}}(x)$ via a continuous trajectory. The forward process progressively perturbs x_0 into a noisy state x_t according to a predefined schedule, formulated as a stochastic differential equation (SDE):

$$\mathrm{d}\boldsymbol{x}_t = \boldsymbol{f}(t, \boldsymbol{x}_t)\,\mathrm{d}t + g(t)\,\mathrm{d}\boldsymbol{w}_t, \tag{1}$$

where $\boldsymbol{w}_t$ is standard Brownian motion, and $\boldsymbol{f}(\cdot)$ and $g(\cdot)$ are drift and diffusion coefficients. This process induces a family of marginals $\{p_t(\boldsymbol{x})\}_{t\in[0,T]}$ that interpolate between the data distribution and a Gaussian prior. Generation reverses this dynamic. Following the score-SDE framework [Song *et al.*, 2021], the reverse-time process is expressed as:

$$\mathrm{d}\boldsymbol{x}_t = \big[\boldsymbol{f}(t, \boldsymbol{x}_t) - g(t)^2 \nabla_{\boldsymbol{x}} \log p_t(\boldsymbol{x}_t)\big]\,\mathrm{d}t + g(t)\,\mathrm{d}\bar{\boldsymbol{w}}_t, \tag{2}$$

where $\nabla_{\boldsymbol{x}} \log p_t(\boldsymbol{x})$ is the time-dependent score function. Alternatively, deterministic trajectories are obtained via the probability flow ordinary differential equation (PF-ODE):

$$\mathrm{d}\boldsymbol{x}_t = \big[\boldsymbol{f}(t, \boldsymbol{x}_t) - \tfrac{1}{2} g(t)^2 \nabla_{\boldsymbol{x}} \log p_t(\boldsymbol{x}_t)\big]\,\mathrm{d}t. \tag{3}$$

Diffusion models typically parameterize the score function via noise prediction $\boldsymbol{\epsilon}_\theta(x_t, t)$, and flow-matching models directly learn a velocity field $\boldsymbol{v}_\theta(x_t, t)$ that specifies the generative dynamics. **Throughout this survey, we adopt a unified view of diffusion and flow-matching models as continuous-time generative processes parameterized by a time-dependent score function, and thus discuss most alignment objectives jointly**. We only distinguish them when practical differences matter, such as when deterministic sampling calls for additional stochasticity in RL-style exploration or when likelihood surrogates differ.

2.2 Alignment Formalization

Let p_{ref} denote a reference generator (e.g., a pretrained model) inducing a distribution over outputs $\boldsymbol{x}_0$ conditioned on y. Given a scalar preference signal $r(\boldsymbol{x}_0, y)$, alignment aims to reshape the model distribution p_θ toward high-reward samples while minimizing distributional drift. This is formalized as a reward-regularized objective:

$$\max_{p_\theta} \mathbb{E}_{\boldsymbol{x}_0 \sim p_\theta}[r(\boldsymbol{x}_0, y)] - \lambda \mathbb{D}_{\text{KL}}(p_\theta \| p_{\text{ref}}), \tag{4}$$

where λ trades off preference improvement against deviation from the reference distribution. This provides a unified perspective: training-time methods optimize model parameters θ to maximize this expected return, whereas inference-time methods utilize $r(\boldsymbol{x}_0, y)$ to steer individual sampling trajectories without modifying the underlying weights. In practice, the feedback $r(\cdot)$ may be provided as scalar scores or pairwise comparisons.

3 Training-Time Alignment

Training-time alignment reshapes diffusion and flow-matching generators by converting preference feedback into learning signals that update model parameters. In this section, we begin with reward modeling for generative alignment (Sec. 3.1), introducing common feedback sources, then we present several representative training-time paradigms. For each paradigm, we present the core objective and training recipe, summarize representative extensions and improvement directions, and conclude with a comparative discussion of trade-offs, limitations, and open questions.

3.1 Reward Signals for Generative Alignment

Alignment methods for diffusion and flow-matching models typically necessitate an evaluator that measures how well a generated sample adheres to a desired behavior under a given condition. Formally, let $r(\boldsymbol{x}_0, y)$ denote a reward signal where y denotes the condition (e.g., a text prompt), and x represents the generated visual output. Depending on the modeling form, $r(\cdot)$ is formulated as a scalar score or as a preference rank between sample pairs.

In practice, preference signals arise from several common sources. (1) **Learned reward models:** Models trained on preference datasets to score or rank multiple $(\boldsymbol{x}_i, y)$ candidates. (2) **Human feedback:** Explicit ratings or pairwise comparisons provided by human annotators. (3) **VLM-as-a-judge:** Large VLMs utilized to generate preference choices and critiques. (4) **Verifiable systems:** Programmatic evaluators providing verifiable signals for objective constraints, such as OCR accuracy for text rendering. (5) **Implicit Preferences:** Pairs constructed without manual labeling, typically leveraging the margin between real and generated samples or between successive iterations of model outputs.

For learned reward models, model is typically parameterized as a critic $r_\phi(x, y)$, and optimized through either pointwise or pairwise supervision. Pointwise supervision treat reward modeling as a regression task, training the critic to predict an absolute score z:

$$\mathcal{L}_{\text{pt}}(\phi) = \mathbb{E}_{(y,\boldsymbol{x},z)\sim\mathcal{D}}\big[\ell\big(r_\phi(\boldsymbol{x}, y),\, z\big)\big], \tag{5}$$

where $\ell(\cdot)$ denotes a standard distance metric. Alternatively, pairwise supervision optimizes for relative preferences. Given a triplet $(y, \boldsymbol{x}^+, \boldsymbol{x}^-)$ where $\boldsymbol{x}^+ \succ \boldsymbol{x}^-$, the Bradley-Terry model specifies:

$$P(\boldsymbol{x}^+ \succ \boldsymbol{x}^- \mid y) = \sigma\big(r_\phi(\boldsymbol{x}^+, y) - r_\phi(\boldsymbol{x}^-, y)\big)\,,$$
$$\mathcal{L}_{\text{pr}}(\phi) = -\mathbb{E}\big[\log P(\boldsymbol{x}^+ \succ \boldsymbol{x}^- \mid y)\big]. \tag{6}$$

Architectural backbones for r_ϕ have evolved into three distinct paradigms. *CLIP-based reward models* [Xu *et al.*, 2023; Kirstain *et al.*, 2023; Wu *et al.*, 2023] leverage pretrained CLIP encoders with lightweight MLP heads. While computationally efficient, the performance is bounded by CLIP's capacity. *VLM-based reward models* [Liu *et al.*, 2025b; Ma *et al.*, 2025c; Wu *et al.*, 2025b] employ stronger VLM backbones to extract scalar rewards. These models represent the dominant choice, as they are generally more accurate and benefit directly from backbone scaling. Recent work also explores Chain-of-Thought (CoT) reasoning [Wang *et*

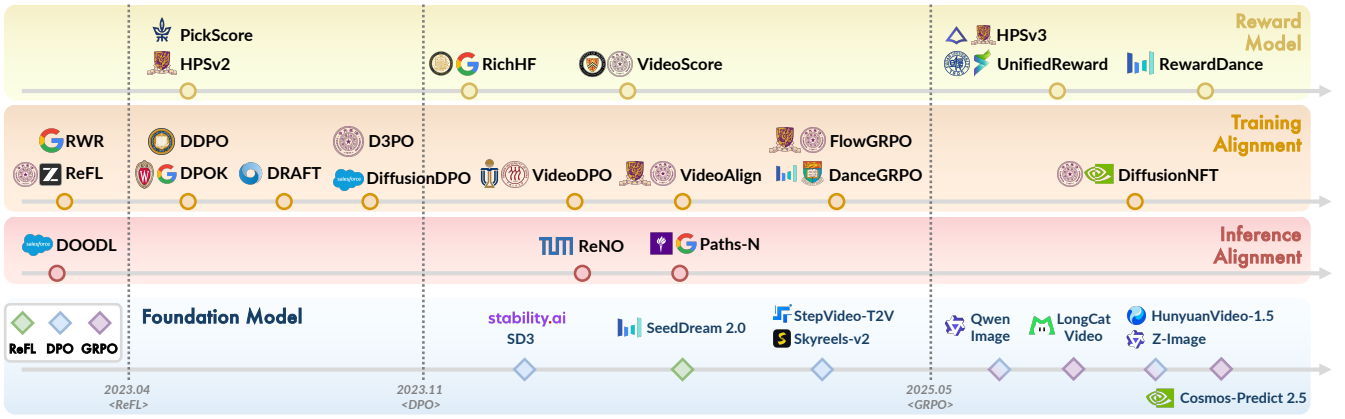


Figure 1: The rising trend of alignment for diffusion and flow-matching models. We summarize representative works on reward modeling, training-time, and inference-time alignment, and highlight several foundation models that have enabled certain alignment techniques at scale.

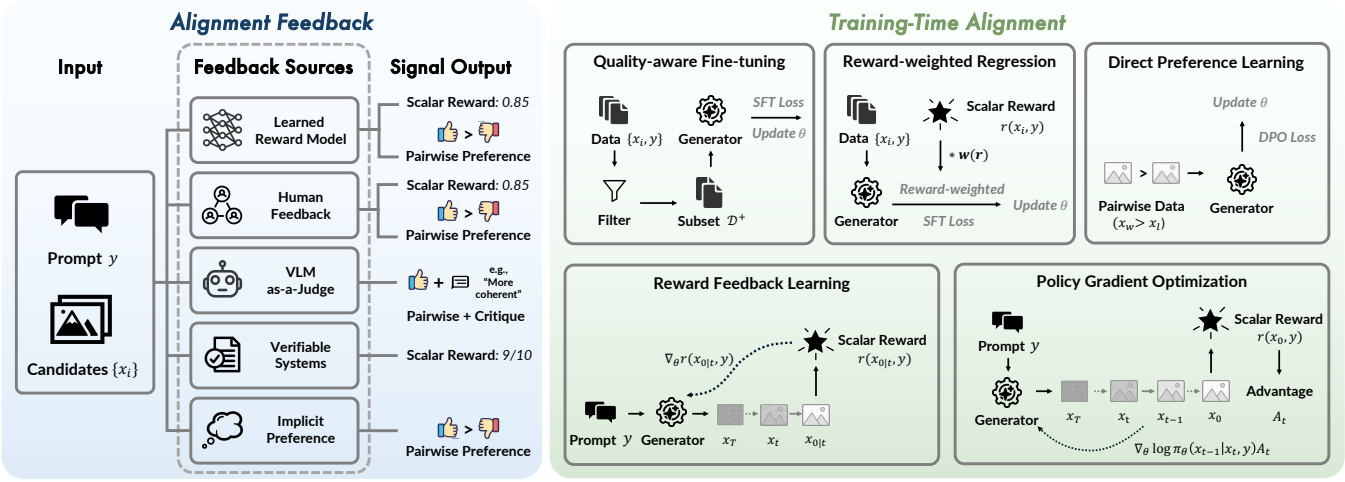


Figure 2: Alignment feedback and training-time alignment paradigms for diffusion and flow-matching models. We summarize representative feedback sources and training-time alignment recipes, showing how different feedback signals are incorporated into optimization pipelines.

al., 2025c; Wu *et al.*, 2025c; Wang *et al.*, 2025b] to improve the judgment ability. Finally, *Diffusion-based reward models* [Zhang *et al.*, 2025; Mi *et al.*, 2025] serve as an emerging alternative that leverages the generator's own latent space for evaluation. By bypassing explicit pixel-space decoding during alignment training, these models offer improved efficiency for online alignment.

3.2 Quality-aware Fine-tuning

We begin with **Quality-aware Fine-tuning (QFT)**, a simple yet widely adopted alignment recipe that improves a generator by refining its training data distribution. Rather than modifying the optimization target, QFT leverages preference signals to *select* high-quality training examples, and then fine-tunes the model using the standard diffusion or flow-matching objective. By filtering the training set via preference signals, the model is encouraged to shift its density toward regions of the output space that better adhere to desired behaviors.

A representative pipeline follows a structured sequence. First, a candidate pool x_i is assembled by collecting either existing training samples or synthetic outputs generated by a base model for a given condition y. These samples are evaluated using an external signal $r(\boldsymbol{x}_i, y)$ derived from feedback systems. A high-quality subset, $\mathcal{D}_{\text{HQ}}$, is subsequently constructed through thresholding or top-K selection. Finally, the generator is fine-tuned on this curated dataset using the same objective as pretraining:

$$\mathcal{L}_{\text{QFT}}(\theta) = \mathbb{E}_{(x_0,y)\sim\mathcal{D}_{\text{HQ}}}\Big[\mathcal{L}_{\text{base}}(\theta; x_0, y)\Big], \tag{7}$$

This approach has become the de facto post-training practice for aligning large-scale foundation models [Wu *et al.*, 2025a; Wan *et al.*, 2025]. After pretraining, models are further fine-tuned on curated high-quality data to improve capability. Beyond its utility in post-training enhancement, this method is also frequently employed to validate the effectiveness and generalization of new reward models [Liang *et al.*, 2024].

Discussion QFT is attractive for its simplicity and stability: it requires no changes to the training pipeline beyond *data selection*, and it can incorporate almost any form of reward or

preference signal. However, its alignment effect is inherently *indirect* and often bounded by the coverage of $\mathcal{D}_{\text{HQ}}$: selection rules may amplify biases in the reward signal, overly prioritizing high-scoring samples while under-exposing the model to low-scoring cases that are crucial for correcting failures. We expect QFT to remain a practical workhorse in large-scale model development; nevertheless, it is typically insufficient, and more targeted alignment techniques are often required on top of QFT to enforce finer-grained preferences and resolve nuanced failure modes.

3.3 Reward-weighted Regression

While QFT reshapes the effective training distribution through explicit data selection, **Reward-weighted Regression (RWR)** incorporates preference signals more directly by *reweighting the learning objective*. RWR can be viewed as a hybrid paradigm between SFT and reinforcement learning. It treats alignment as a weighted maximum likelihood estimation problem, where samples with higher scalar reward contribute more strongly to the parameter update.

The classical formulation of RWR involves minimizing a weighted log-likelihood objective [Peng *et al.*, 2019], which optimizes a lower bound on the expected reward. Given a condition y and a sample x, the objective is formulated as:

$$\mathcal{L}_{\text{RWR}}(\theta) = -\mathbb{E}_{\boldsymbol{x},y}\left[\exp\left(\frac{r(\boldsymbol{x},y)}{\beta}\right)\log \pi_\theta(\boldsymbol{x}|y)\right], \quad (8)$$

where $r(\boldsymbol{x}, y)$ denotes the scalar reward and β controlls the concentration of the reward-weighted distribution. In diffusion and flow-matching models, Eq. 8 is typically implemented by scaling the standard denoising or velocity prediction loss with per-sample reward weights [Lee *et al.*, 2023].

A practical limitation of RWR is that it is often implemented with a fixed dataset, which can become increasingly misaligned as the model updates. To alleviate this issue, recent extensions such as ORW [Fan *et al.*, 2025] incorporate *online sampling* and introduce Wasserstein-based regularization to stabilize iterative updates under distribution shift.

Discussion Generally, RWR turns preference alignment into *reward-weighted supervised learning*, applying sample-level weights to guide denoising updates, and thus requires a scalar reward. On fixed offline datasets, RWR can exploit reward signals more continuously than QFT, but is often less flexible in practice, especially for multi-dimensional preferences. Online variants partially mitigate off-policy drift, yet typically make less effective use of collected samples and rewards than on-policy methods such as ReFL or GRPO, with comparatively weaker regularization against reward hacking. As a result, RWR remains a lightweight and useful baseline, but is less commonly adopted as a general-purpose solution for large-scale alignment.

3.4 Reward Feedback Learning

Reward feedback learning (ReFL) [Xu *et al.*, 2023] aligns diffusion models by directly optimizing a differentiable preference signal. Given a differentiable reward model r_ϕ, ReFL estimates the reward based on a predicted clean image at a randomly sampled denoising step t. To keep training tractable, the reward gradients are backpropagated through only a single denoising step. This single-step truncation focuses optimization on the terminal stages of the generation process, where the latent samples exhibit well-defined semantic structures. Subsequent works have extended this paradigm to **accommodate longer sampling chains** [Clark *et al.*, 2023; Prabhudesai *et al.*, 2023] or **adapt it for video diffusion models** [Yuan *et al.*, 2024; Prabhudesai *et al.*, 2024]. The generalized objective is formulated as:

$$\mathcal{L}_{\text{ReFL}}(\theta) = -\,\mathbb{E}_{y\sim\mathcal{D}}\,\mathbb{E}_{\boldsymbol{\epsilon}\sim\mathcal{N}(0,I)}\big[r_\phi(g_\theta(\boldsymbol{\epsilon},y),y)\big], \quad (9)$$

where $g_\theta(\boldsymbol{\epsilon}, y)$ denotes the (possibly truncated) generation.

Discussion Generally, ReFL performs alignment by *back-propagating reward gradients* into the denoising model, making it an efficient route to improve differentiable scores. Compared to reward-weighted objectives such as RWR, ReFL can impose more direct pressure toward high-reward behaviors, but its scalability is often limited by the memory and compute cost of differentiating through reward models and denoising steps, which in practice encourages short-horizon truncation. This restriction can weaken the model's ability to correct early-stage structural errors and also increase vulnerability to reward hacking under imperfect rewards, where optimization improves numerical scores at the expense of perceptual fidelity or diversity. Mitigations such as reward scaling [Wu *et al.*, 2025b] can alleviate over-optimization, and ReFL-style post-training remains an important component in several state-of-the-art foundation models [Gong *et al.*, 2025; Gao *et al.*, 2025].

3.5 Policy Gradient Optimization

Policy-gradient methods explicitly target the expected return of the sampling trajectory by treating generation as a sequential decision-making process. By operating on on-policy rollouts and optimizing a trajectory-level policy objective, these approaches utilize the sparse terminal feedback to update the entire denoising policy while maintaining trust-region control. This line of research therefore reformulates the sampling procedure as a Markov Decision Process (MDP) to perform direct, trajectory-level reinforcement learning.

MDP formulation To apply RL, the denoising process of diffusion models is typically modeled as a T-horizon MDP $\mathcal{M} = (\mathcal{S}, \mathcal{A}, P, R, \gamma)$ [Black *et al.*, 2023; Fan *et al.*, 2023], which defines the state s_t as the current noisy latent and the action a_t as the next denoised latent:

$$\begin{aligned}
&\boldsymbol{s}_t \triangleq (y, t, \boldsymbol{x}_t), \quad \mathbf{a}_t \triangleq \boldsymbol{x}_{t-1} \\
&\pi(\boldsymbol{a}_t|\boldsymbol{s}_t) \triangleq p_\theta(\boldsymbol{x}_{t-1}|\boldsymbol{x}_t, y), \;\; P(\boldsymbol{s}_{t+1}|\boldsymbol{s}_t, \boldsymbol{a}_t) \triangleq (\delta_y, \delta_{t-1}, \delta_{\boldsymbol{x}_{t-1}}) \\
&P_0(\boldsymbol{s}_0) \triangleq (p(y), \delta_T, \mathcal{N}(\mathbf{0}, \mathbf{I})), \;\; R(\boldsymbol{s}_t, \boldsymbol{a}_t) \triangleq r(\boldsymbol{x}_0, y)\cdot\mathbb{I}(t=0).
\end{aligned} \quad (10)$$

The reward is defined only on the final decoded output $\boldsymbol{x}_0$.

For flow-matching models, sampling is implemented as a deterministic ODE, which lacks the stochastic exploration needed by standard on-policy RL. A common remedy [Liu *et al.*, 2025a] is to construct an SDE counterpart that preserves the marginal distribution while injecting stochasticity:

$$\mathrm{d}\boldsymbol{x}_t = \left(\boldsymbol{v}_\theta(\boldsymbol{x}_t) - \frac{\sigma_t^2}{2}\nabla \log p_t(\boldsymbol{x}_t)\right)\mathrm{d}t + \sigma_t \mathrm{d}\boldsymbol{w}, \quad (11)$$

where $\boldsymbol{v}_\theta$ is the learned velocity field and σ_t modulates exploration. Discretizing Eq. 11 yields a Gaussian transition:

$$\boldsymbol{x}_{t+\Delta t} = \boldsymbol{x}_t + \Big[\boldsymbol{v}_\theta(\boldsymbol{x}_t, t) - \frac{\sigma_t^2}{2}\nabla \log p_t(\boldsymbol{x}_t)\Big]\Delta t + \sigma_t\sqrt{\Delta t}\,\epsilon. \tag{12}$$

These constructions make it possible to define per-step likelihood ratios and apply policy-gradient objectives to both diffusion and flow-matching generators within a unified RL view.

Proximal Policy Optimization Early works established the feasibility of Proximal Policy Optimization (PPO) for diffusion alignment by viewing the reverse denoising chain as a sequence of policy actions. DDPO [Black *et al.*, 2023] and DPOK [Fan *et al.*, 2023] demonstrate that text-to-image generators can be fine-tuned to maximize non-differentiable rewards through on-policy rollouts. Given a trajectory $\tau = \{\mathbf{x}_T, \ldots, \mathbf{x}_0\}$, PPO optimizes a clipped surrogate objective:

$$\mathcal{L}_{\text{PPO}}(\theta) = \mathbb{E}_{\tau\sim\pi_{\text{old}}}\left[\sum_{t=1}^{T}\min\Big(\rho_t(\theta)\hat{A}_t, \text{clip}(\rho_t(\theta), 1\pm\epsilon)\hat{A}_t\Big)\right], \tag{13}$$

where $\rho_t(\theta) = \frac{\pi_\theta(\mathbf{x}_{t-1}|\mathbf{x}_t,\mathbf{c})}{\pi_{\text{old}}(\mathbf{x}_{t-1}|\mathbf{x}_t,\mathbf{c})}$ is the importance sampling ratio and ϵ controls the trust-region. The advantage $\hat{A}_t$ is typically estimated with a learned or defined value baseline $V_\psi(\mathbf{s}_t)$,

$$\hat{A}_t = r(\boldsymbol{x}_0, y) - V_\psi(\mathbf{s}_t). \tag{14}$$

In practice, DPOK further incorporates KL regularization against a reference policy to prevent excessive deviation from the pretrained model, and some works explore diversity-preserving regularizers to alleviate reward-driven collapse [Black *et al.*, 2023].

Group Relative Policy Optimization A practical bottleneck of PPO-style alignment is its reliance on accurate advantage estimation, which couples training to a critic. For high-dimensional diffusion generators with sparse terminal rewards, training a reliable value function can be difficult and may introduce instability. Group Relative Policy Optimization (GRPO) [Shao *et al.*, 2024] has recently emerged as a compelling alternative. By removing the critic and estimating relative advantages from a group of sampled outputs, GRPO significantly simplifies training and improves robustness.

Representative methods such as Flow-GRPO [Liu *et al.*, 2025a] and DanceGRPO [Xue *et al.*, 2025] validate this paradigm across diverse objectives including preference alignment, compositional generation, and complex text rendering. Given a prompt $\mathbf{y}$, GRPO samples a group of G trajectories $\{\tau_1, \tau_2, \ldots, \tau_G\}$ from the old policy π_{old} and computes normalized advantages from the group statistics:

$$\hat{A}_t^i = \frac{R(\boldsymbol{x}_0^i, \mathbf{y}) - \text{mean}(\{R(\boldsymbol{x}_0^i, \mathbf{y})\}_{i=1}^G)}{\text{std}(\{R(\boldsymbol{x}_0^i, \boldsymbol{c})\}_{i=1}^G)}. \tag{15}$$

The policy is then optimized with a clipped objective and an explicit KL constraint to the reference policy π_{ref}:

$$\mathcal{L}_{\text{GRPO}}(\theta) = \mathbb{E}_{\mathbf{y},\tau_i\sim\pi_{\text{old}}}\Bigg[\frac{1}{G}\sum_{i=1}^{G}\sum_{t=1}^{T}\Big(\min\big(\rho_t^i(\theta)A_i, \text{clip}(\rho_t^i(\theta), 1-\epsilon, 1+\epsilon)A_i\big) - \beta\mathbb{D}_{\text{KL}}(\pi_\theta|\pi_{\text{ref}})\Big)\Bigg], \tag{16}$$

where $\rho_t^i(\theta)$ denotes the per-step importance ratio.

Further improvements Despite strong empirical performance, trajectory-level on-policy optimization for visual generators continues to face challenges in **efficiency, robustness, and credit assignment**. To improve **computational efficiency**, which is primarily limited by the necessity of calculating policy gradients at every discrete denoising step, recent methods like TempFlow-GRPO [He *et al.*, 2025] and MixGRPO [Li *et al.*, 2025a] employ hybrid sampling strategies. They restrict exploration and gradient updates to a sparse subset of discrete steps, using deterministic paths for the remainder of the trajectory. Robustness issues are often dominated by **reward hacking**, where reward scores improve while perceptual quality or diversity degrades. Recent work mitigates such failures through improved trust-region control (e.g., renormalized ratios in GRPO-Guard [Wang *et al.*, 2025a]), better exploration via SDE design [Wang and Yu, 2025] and sampling strategies [Li *et al.*, 2025c; Sheng *et al.*, 2025], as well as refined initial noise distributions [Yu *et al.*, 2025]. Finally, **credit assignment** remains difficult because rewards are typically terminal-only. Partial remedies include timestep-wise reweighting [He *et al.*, 2025] and chunk-wise or denser rewards [Luo *et al.*, 2025] to provide more informative supervision along the trajectory.

Discussion Generally, policy-gradient alignment treats diffusion and flow sampling as *on-policy trajectory optimization*, providing a principled framework to exploit non-differentiable judges and verifiable rewards using explicit trust-region control. Compared to other on-policy methods such as online-RWR or ReFL, PPO and GRPO can propagate terminal feedback to the entire denoising process and often yield stronger gains when reward signals are reliable. However, this comes with higher sampling cost and engineering complexity, and diffusion-specific numerical issues (e.g., unstable importance ratios calculation and clipping) can exacerbate instability and reward hacking. An open question is how to make on-policy alignment both compute-efficient and robust at scale, while improving temporal credit assignment without sacrificing diversity or visual fidelity. Nonetheless, this paradigm shows strong promise for complex generative refinement and for fully leveraging rich reward signals, and has been adopted in the large-scale post-training of recent foundation models via GRPO-style optimization [Wu *et al.*, 2025a; Ali *et al.*, 2025]. We expect its importance to grow as reward models and judges continue to improve.

3.6 Direct Preference Learning

In contrast to other reward-based pipelines, Direct Preference Learning (DPL) updates the generator *directly from pairwise comparisons*, without requiring an explicit reward model or a multi-stage RL loop. Given a condition y and a preference pair (x^+, x^-) with $x^+ \succ x^-$, DPL encourages the generator to assign higher likelihood to the preferred sample x^+, thereby reinforcing desired behaviors while suppressing undesirable ones. This comparison-only paradigm makes DPL particularly *flexible* in practice, since pairwise judgments are often easier to collect and more stable than absolute reward calibration.

Direct Preference Optimization A milestone work is Direct Preference Optimization (DPO) [Rafailov *et al.*, 2023], which can be derived from the KL-regularized reward maximization objective in Eq. 4. Under this view, the optimal policy admits a closed form $p^\star(x \mid y)$, allowing the reward to be reparameterized by the policy ratio, yielding an *implicit reward* $r(x, y) = \beta(\log p^\star(x \mid y) - \log p_{\mathrm{ref}}(x \mid y)) + C(y)$. Assuming a Bradley-Terry preference model, DPO reduces optimization to a contrastive logistic loss that compares likelihood ratios under the current policy and a fixed reference policy:

$$\mathcal{L}_{\mathrm{DPO}}(\theta) = -\,\mathbb{E}\left[\log \sigma\left(\beta \log \frac{p_\theta(x^+ \mid y)}{p_{\mathrm{ref}}(x^+ \mid y)} - \beta \log \frac{p_\theta(x^- \mid y)}{p_{\mathrm{ref}}(x^- \mid y)}\right)\right]. \tag{17}$$

Here β controls the anchoring strength to the reference policy p_{ref}, with a larger β yielding more conservative updates.

Applying Eq. 17 to diffusion models is non-trivial because exact likelihoods are intractable. DiffusionDPO [Wallace *et al.*, 2024] resolves this by replacing $\log p_\theta(x \mid y)$ with a tractable denoising-based surrogate:

$$\begin{aligned}\mathcal{L}_{\mathrm{DiffusionDPO}}(\theta) = -\,\mathbb{E}\Big[\log \sigma\Big(-\beta\, T\, \omega(\lambda_t)\Big(&\\ \big\|\epsilon^w - \epsilon_\theta(x_t^w, t, y)\big\|_2^2 - \big\|\epsilon^w - \epsilon_{\mathrm{ref}}(x_t^w, t, y)\big\|_2^2&\\ - \Big(\big\|\epsilon^l - \epsilon_\theta(x_t^l, t, y)\big\|_2^2 - \big\|\epsilon^l - \epsilon_{\mathrm{ref}}(x_t^l, t, y)\big\|_2^2\Big)\Big)\Big)\Big].&\end{aligned} \tag{18}$$

Formally, this objective compares the denoising errors of the current model and the reference model along the x^+ and x^- trajectories. Subsequent works generalize this idea to flow matching models by replacing the ϵ-prediction surrogate with a v-prediction surrogate [Liu *et al.*, 2025b].

Further Improvements Diffusion-native DPL variants mainly differ in *how preference supervision is defined* and *where it is applied* along the noisy trajectory. A key challenge is **preference inconsistency across noise levels**: a single global pairwise label may not remain valid at intermediate timesteps, since forward diffusion can corrupt visual details and weaken or even reverse preference signals. Step-aware methods such as SPO [Liang *et al.*, 2025] therefore incorporate timestep-dependent supervision, while [Yang *et al.*, 2024] introduces temporal discounting and reweighting to emphasize influential stages that shape global structure and prompt fidelity. Beyond reweighting across timesteps, preference signals can be made **denser and more localized**: PatchDPO [Huang *et al.*, 2025] replaces whole-image comparisons with local region preferences, and DenseDPO [Wu *et al.*, 2025d] leverages short temporal segments for more informative video supervision. Moreover, ranking-style objectives [Karthik *et al.*, 2025; Bai *et al.*, 2025] exploit multiple candidates per prompt by learning from listwise orders rather than only pairwise labels. Another practical limitation is **distribution shift**: preference pairs collected from a fixed generator become increasingly off-policy as the model improves, motivating online DPO-style training that alternates between sampling from the current model and optimizing on freshly collected comparisons [Zhang *et al.*, 2024]. Finally, preference relations can also be **constructed implicitly** without explicit human-labeled pairs, for example by treating real data as "chosen" and model outputs as "rejected" to encourage realism [Na *et al.*, 2025; Cheng *et al.*, 2025], by adopting discriminative training objectives such as DDO [Zheng *et al.*, 2025b], or by exploiting self-improvement signals such as the discrepancy between current parameters and an EMA teacher in DiffusionNFT [Zheng *et al.*, 2025a].

Discussion Generally, DPL makes preference alignment *comparison-driven and reward-free*: it relies only *on pairwise judgments*, avoiding explicit reward modeling and the complexity of on-policy RL. This simplicity yields strong practical advantages, including stable optimization, easy integration into standard fine-tuning pipelines, and broad applicability whenever preference pairs can be collected at scale. DPL-style objectives have become a widely adopted post-training recipe for large-scale visual foundation models [Esser *et al.*, 2024; Wu *et al.*, 2025a; Ma *et al.*, 2025a]. However, DPL also inherits the limitations of the preference data it learns from: biases in pair construction can be amplified, and offline optimization is vulnerable to off-policy distribution shift as the generator improves. We expect DPL to remain a practical alignment primitive, while continued progress will likely depend on better pair generation, step-aware supervision, and online data collection strategies that improve robustness without sacrificing diversity.

4 Inference-Time Alignment

While training-time alignment reshapes the generator by updating model parameters, it can be computationally intensive and less flexible when preferences are diverse, evolving, or task-specific. Inference-time alignment offers a complementary alternative: it keeps the base generator fixed and intervenes only in the sampling procedure. This paradigm views alignment as *steering or searching* over the denoising trajectory, using external feedback to bias generation toward high-reward regions without weight updates.

We categorize inference-time alignment into primary frameworks: (i) **gradient guidance**, which perturbs the sampling trajectory using differentiable objectives; (ii) **search and selection**, which allocates extra test-time compute to explore candidate trajectories and select high-reward samples; and (iii) **conditioning steering**, which optimizes external conditions like prompts or embeddings.

Gradient Guidance A direct route to inference-time alignment is to inject gradient-based steering terms into the sampling dynamics, extending the idea of classifier guidance to more general objectives. Universal guidance [Bansal *et al.*, 2023] generalizes this principle by allowing arbitrary differentiable reward functions and using their gradients to guide the denoising trajectory. Similarly, DOODL [Wallace *et al.*, 2023] enables plug-and-play alignment by optimizing diffusion latents with gradients derived from pre-trained pixel-space classifiers. A related variant shifts the intervention point from intermediate denoising states to the initial noise. Instead of applying per-step perturbations, these methods optimize the starting noise through reward gradients and inversion-based initialization [Eyring *et al.*, 2024; Qi *et al.*, 2024].

Search and Selection Beyond local gradient-based steering, inference-time alignment can also be formulated as *a discrete search problem* that utilizes extra test-time compute to explore a larger solution space. The simplest form is best-of-N sampling, which draws multiple candidates and selects the one with the highest terminal reward. More structured strategies further improve exploration by introducing beam search [Oshima *et al.*, 2025], tree search [Guo *et al.*, 2025], or trajectory refinement procedures that iteratively improve candidate paths [Ma *et al.*, 2025b]. Complementary lines of work [Zhou *et al.*, 2025] aim to identify *high-reward initialization regions* by learning predictors or heuristics that locate "golden noise" for a given condition, thereby increasing the hit rate of high-quality samples with limited search budget.

Conditioning Steering The third family achieves alignment by optimizing the *conditioning inputs* rather than the sampling dynamics. ReNeg [Li *et al.*, 2025b] learns compact negative embeddings that suppress undesired behaviors, offering a lightweight plug-in control mechanism. At the prompt level, recent works leverage preference-aligned LLMs to rewrite user instructions into more effective prompts [Ji *et al.*, 2025], or establish closed-loop refinement where generated outputs are evaluated and the prompt is iteratively updated based on feedback [Lee and Ye, 2025].

Discussion Inference-time alignment is attractive for its *zero-weight-update* flexibility. It can rapidly incorporate new objectives and naturally supports test-time scaling by spending additional compute. However, its benefits come with a clear cost in per-sample latency, and its effectiveness is ultimately bounded by the reliability of the external evaluator. Imperfect rewards may induce over-optimization, leading to degraded visual fidelity or reduced diversity. Nonetheless, due to its flexibility and low overhead, inference-time alignment can serve as a cost-effective complement or alternative to training-time alignment, and can be useful for rapidly stress-testing reward models, probing reward reliability, estimating alignment potential, and diagnosing possible reward-hacking patterns before expensive training.

5 Challenges and Opportunities

While each alignment paradigm introduces its own technical bottlenecks, many challenges are shared across methods. In this section, we highlight several general-purpose issues in practice and discuss opportunities for future progress.

Reward Robustness during Alignment Alignment is an inherently dynamic process. A reward model that appears reliable on static benchmarks can become increasingly fragile as the generative policy drifts into off-policy regions. During optimization, the generative model actively searches for the flaws of the proxy signal, turning minor reward imperfections into dominant learning shortcuts and producing misleading optimization signals. This calls for more robust evaluation protocols for reward models themselves. Beyond correlation with human rankings, it is crucial to assess *adversarial robustness* and *out-of-distribution reliability*, i.e., whether the reward can distinguish genuinely high-quality samples from those specifically optimized to exploit it. Potential strategies include *ensemble-based rewards* that reduce single-model blind spots, and *uncertainty-aware scoring* that downweights updates when inputs fall into low-confidence regions of the reward model.

Reward Hacking and Over-Optimization. Reward Hacking in visual generation occurs when a policy exploits the imperfections of a proxy reward model to achieve high numerical scores at the expense of visual fidelity or diversity. Beyond improving reward robustness, algorithmic safeguards are often needed to prevent over-optimization. Key strategies include *incorporating KL-divergence regularization* and *trust-region constraints*, such as the importance-ratio clipping used in PPO and GRPO, to anchor the update to the pre-trained prior. Another practical consideration is that different noise levels encode different semantic factors, making *timestep-aware weighting* a natural tool to focus updates on the most influential stages of the trajectory. Data and sampling choices also play a critical role: offline alignment can benefit from careful pair filtering and bias control, while online alignment typically requires *high-diversity rollouts* to avoid collapsing onto narrow reward modes. Finally, sampling-time safeguards, such as conservative filtering or rejection heuristics on extremely high-reward samples, can provide an additional layer of protection against reward-induced artifacts.

Structured Environments In large language models, reinforcement learning has achieved significant success in structured environments such as coding and mathematics, where objectives are verifiable and the environment provides clear, deterministic feedback. These settings enable RL to move beyond mere imitation and improve general reasoning capabilities. In contrast, most current alignment for visual generation remains dominated by *subjective preference optimization* (e.g., aesthetics), whose supervision is often noisy and rarely generalizes to complex problem-solving or logical reasoning. A promising opportunity is to build structured "visual environments" where states, actions, and rewards are defined by *verifiable constraints*, such as geometric correctness, physical consistency, temporal causality, or multimodal reasoning checks. By shifting alignment from subjective scoring toward objective verification, these environments could enable more reliable credit assignment and unlock a richer frontier for post-training visual foundation models.

6 Conclusion

This survey reviews alignment for diffusion and flow-matching generators through a diffusion-native perspective, focusing on how preference feedback reshapes the denoising trajectory. We have proposed a two-level taxonomy separating training-time and inference-time alignment, and further organized training-time methods into five paradigms: QFT, RWR, ReFL, policy-gradient optimization, and DPL. For each paradigm, we have distilled the core objective, representative extensions, and key trade-offs, and summarized diffusion-specific challenges. We expect this survey to provide a reference for systematic comparison and inspire future research to explore the potential of alignment in diffusion models.

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